FRONTIER SYNTHESIS · SPECULATIVE ARCHITECTURE

The Post-Bandwidth Architecture

A Physics-Native Compute Fabric That Removes Data Movement Instead of Speeding It Up

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A Note on What This Document Is

This is a speculative architecture built by combining two independent, early-stage physical computing research lines that are each real and citable as of mid-2026, but that no organization has combined into one system. Unlike a roadmap extension of current accelerator design, this document deliberately abandons the assumption underneath every current high-bandwidth architecture — including the one this paper's predecessor document proposed — that computation requires moving data from storage to a separate processing unit. Both substrates described here compute in place, using physics, instead of moving data to logic.

This is further from deployment than the optical-interconnect roadmap discussed previously. It is presented as a paradigm bet with explicit failure modes, not a near-term engineering plan.

LAB-PROVEN	Demonstrated in working silicon or bench prototypes; not yet productized or at scale.
EARLY TAPE-OUT	First-generation ASIC exists and has been fabricated; not yet commercially deployed.
OPEN RESEARCH	Physically plausible and actively studied; no working device at the scale described.

Executive Summary

Every high-bandwidth architecture in production today — and the optical-interconnect synthesis this paper's predecessor proposed — shares one unexamined assumption: that a computation requires fetching data from where it is stored, delivering it to a separate arithmetic unit, and writing a result back. Making that delivery faster, shorter, or more reconfigurable is a linear improvement with a hard ceiling, because the delivery itself has an irreducible physical cost. The only way to get an exponential result is to stop paying that cost for as many operations as possible.

Two independent physical computing research lines, both active in 2025–2026, do exactly that for different classes of operations. Photonic in-memory tensor cores store neural network weights as the physical state of an optical material and perform multiply-accumulate as a side effect of light passing through that state — the arithmetic happens where the weight already sits, at the speed of light, with no fetch step. Thermodynamic computing chips solve sampling, optimization, and probabilistic inference problems by letting a physical circuit relax toward a natural equilibrium and reading that equilibrium as the answer, rather than computing it through a sequence of digital steps. Both approaches have working silicon in 2025–2026; neither has been combined with the other, and neither has been proven at model scale.

The architecture proposed here, the Post-Bandwidth Architecture, assigns each class of AI workload operator to whichever substrate computes it without moving data, and reserves a conventional digital control plane only for the inherently sequential and symbolic work — tokenization, orchestration, control flow — that neither physics-native substrate is suited to. The result is a system where the interconnect only has to carry the comparatively small handoff signal between substrate regions, not the full volume of weights and activations a conventional accelerator moves for every operation. This is why the bet is exponential rather than linear: a faster pipe is bounded by the physics of transmission; a system that stops needing to transmit scales with how much of the workload can be moved onto a substrate that computes in place, which grows as these two research lines mature independently.

1. The Assumption Every Prior Architecture Shares

Hybrid HBM/SRAM memory, optical chiplets, and reconfigurable optical topology — the three pillars combined in the previous synthesis — all reduce the cost of moving data between separate memory and compute. None of them ask whether that movement needs to happen at all. This is worth stating plainly: those technologies are real, they are shipping or close to it, and they will meaningfully improve the systems that adopt them. They are not, however, a different kind of bet than the HBM/SRAM hybrid architecture that came before them. Each generation makes the pipe wider or shorter. None removes the pipe.

The two substrates below are each, independently, an attempt to remove the pipe for a specific class of computation — not by making data transfer faster, but by making the storage location and the computing element the same physical object.

2. Two Substrates That Compute Without Moving Data

2.1 Photonic in-memory tensor cores — the deterministic substrate

A photonic tensor core encodes a weight matrix as the physical state of a non-volatile optical material, typically a phase-change material, embedded directly in the optical path. When light carrying an input signal passes through that material, the interaction performs a multiply; combining many such interactions across wavelength-multiplexed channels performs the accumulate. The weight never leaves the device and the input never has to be fetched into a separate arithmetic unit — the multiply-accumulate is a physical side effect of transmission, not a subsequent step after transmission. Demonstrations of this approach report throughput on the order of 10^{12} multiply-accumulate operations per second, and co-locating analog memory directly with photonic compute has been shown to cut power by roughly 26-fold relative to a conventional SRAM-plus-digital-converter design, specifically by removing the repeated analog-to-digital and digital-to-analog conversions that a separate-memory architecture requires.

This substrate is naturally suited to the dense linear algebra that dominates transformer inference and training — attention projections and feed-forward layers — because those operations are fixed matrix multiplications against a weight set that changes slowly relative to how often it is used. Current limitations are precision (roughly 7-bit demonstrated) and the endurance of the phase-change material under sustained, repeated switching, which matters more for training than for inference.

2.2 Thermodynamic computing — the stochastic substrate

A thermodynamic or probabilistic computing chip does not execute a sequence of logic operations to compute a result. It sets up a physical circuit — built from coupled analog elements whose natural noise and relaxation dynamics are the computing resource, not something to be filtered out — and lets that circuit settle toward the physical state that represents the answer. This is a natural fit for exactly the operations that are expensive on digital hardware precisely because they are stochastic: sampling from a distribution, denoising steps in a diffusion model, combinatorial search of the kind that appears in mixture-of-experts routing, and Bayesian uncertainty quantification. Normal Computing's CN101, taped out in 2025 and targeting linear algebra and probabilistic sampling primitives for AI and scientific workloads, is the first chip built specifically around this principle; a second-generation chip aimed at diffusion-model workloads was targeted for 2026. Reported efficiency gains for suitable workloads run as high as three orders of magnitude relative to GPU baselines, though these figures come from the vendor and from narrow benchmark primitives rather than full production models, and should be treated accordingly.

The core limitation is scope: thermodynamic hardware is not a general-purpose replacement for digital logic, and is explicitly unsuited to tasks that require exact, repeatable results — which is precisely why it is a poor fit for the deterministic matrix multiplies the photonic substrate handles, and a good fit for the stochastic operations that substrate does not.

3. Why Combine Them Rather Than Choose One

The two substrates are not competing solutions to the same problem; they are complementary solutions to two different halves of what a modern AI workload actually contains. A transformer-based model's cost is dominated by deterministic dense linear algebra — the photonic substrate's strength. A diffusion model, an agentic system doing uncertainty-aware planning, or a mixture-of-experts model choosing which expert to route to all contain a meaningful share of stochastic, sampling-shaped, or combinatorial computation — the thermodynamic substrate's strength. A system built on only one substrate would still need a conventional digital fallback for the operations the other substrate handles natively, which reintroduces the data-movement cost the whole architecture is trying to remove.

The table below maps common AI workload operators to the substrate that can execute them without a fetch step, and is explicit about how mature each mapping is.

Workload operator	Native substrate	Why data movement disappears	Maturity
Dense matmul (attention/FFN projections)	Photonic in-memory tensor core	Weights held as optical material state; light multiplies by physically passing through them	LAB-PROVEN
Sampling / diffusion denoising steps	Thermodynamic substrate	Answer is the equilibrium state a physical circuit settles into, not a computed sequence of steps	EARLY TAPE-OUT
Discrete search / MoE routing / combinatorial choice	Thermodynamic substrate (Ising-style)	Routing decision emerges from energy minimization across coupled physical states	LAB-PROVEN
Uncertainty quantification / Bayesian layers	Thermodynamic substrate	Noise is the computing resource, not an error to average out over repeated digital passes	OPEN RESEARCH (at model scale)
Tokenization, control flow, orchestration	Thin digital control plane	Inherently sequential/symbolic; kept conventional by design, not a target for physics-native compute	LAB-PROVEN

Table 1. Operator-to-substrate mapping in the Post-Bandwidth Architecture.

4. System Architecture

The Post-Bandwidth Architecture is organized as three tiers rather than the memory-hierarchy-plus-interconnect model conventional accelerators use:

- Photonic tensor tiles: physically hold model weights as optical material state and execute dense linear algebra in place, organized so that a full attention or feed-forward layer's weights live on one tile rather than being streamed in per token.

- Thermodynamic tiles: physically encode the probabilistic or combinatorial structure of a sampling, routing, or uncertainty-estimation step and return an equilibrium state as the result, rather than a value computed through digital iteration.
- A thin digital control plane: handles tokenization, sequencing, and orchestration across tiles — the genuinely sequential and symbolic parts of running a model — and is deliberately kept conventional rather than forced onto a physics-native substrate it is not suited for.

The interconnect's job shrinks correspondingly. In a conventional accelerator, the fabric carries the full volume of weights and activations for every operation. In this architecture, weights never leave the tile that holds them, and most activations are consumed by the same tile that produces the next operation in a layer. What remains for the interconnect to carry is the handoff signal at the boundary between a deterministic region and a stochastic region — for example, the reduced representation passed from an attention block's photonic tile to a routing decision's thermodynamic tile — which is a small fraction of the raw tensor traffic a conventional system would move for the same computation. This is the specific mechanism behind the exponential framing: interconnect bandwidth requirements scale with the number and size of substrate boundaries in a workload graph, not with the workload's total parameter count or FLOP count, and the former grows far more slowly than the latter as models scale up.

5. Research Roadmap

Each horizon below has to be validated independently; nothing after H1 is meaningful if H1's precision and endurance blockers are not resolved first.

Horizon	Milestone	Key blocker to retire	Signal it worked
H1 (1–2 yrs)	Photonic tensor tiles handle one full transformer layer's linear algebra end to end	Analog precision (currently ~7-bit) and phase-change material endurance under sustained training loads	Inference accuracy parity with digital baseline on a real model layer
H2 (2–3 yrs)	Thermodynamic tiles handle diffusion sampling or MoE routing in a production-scale model	Reproducibility: stochastic hardware must be tamed enough for auditable, debuggable production inference	Reported efficiency gains hold on a real workload, not just benchmark primitives
H3 (3–6 yrs)	A compiler maps a model graph onto photonic, thermodynamic, and digital tiles automatically	No existing software stack partitions a workload across fundamentally different physical computing paradigms	A single model trains or serves across all three substrates without hand-written per-layer placement
H4 (6+ yrs, speculative)	Handoff fabric between substrates becomes the only remaining bandwidth-bound layer in the system	Whether inter-substrate handoff traffic actually stays small as model and context size grow	System-level throughput scales with model size faster than interconnect bandwidth does

Table 2. Horizon-by-horizon blockers and success signals.

6. Where This Could Fail

- Precision and reproducibility: analog photonic compute currently tops out around 7-bit precision, and thermodynamic hardware is explicitly stochastic by design; production AI systems that need exact, auditable, reproducible outputs may not tolerate either limitation without a costly error-correction or verification layer that reintroduces the digital overhead this architecture is trying to remove.
- Narrow benchmark claims: the efficiency figures for both substrates — 26-fold power reduction, up to three orders of magnitude energy efficiency — come from primitive-level benchmarks or vendor-reported results on narrow workloads, not full production models. There is no guarantee these gains survive contact with a real, heterogeneous, multi-layer AI system.
- No software stack exists: compiling a model graph across three fundamentally different physical computing paradigms — deterministic photonic, stochastic thermodynamic, and conventional digital — is an open research problem with no existing tooling, closer to heterogeneous quantum-classical compilation than to anything in today's ML compiler stacks.
- The handoff fabric could still become the bottleneck: the entire exponential argument rests on inter-substrate handoff traffic staying small as models scale. If future model architectures turn out to need far more cross-substrate communication than current transformer and diffusion designs do, the system degrades back toward the same bandwidth-bound regime this architecture is trying to escape.
- Two immature manufacturing disciplines instead of one: photonic PCM devices and thermodynamic ASICs each have independent yield, endurance, and packaging challenges; combining them multiplies integration risk rather than sharing it.

7. Conclusion

An exponential bet, honestly stated, is a bet that a specific assumption underlying the current paradigm is unnecessary — not a bet that the current paradigm can be made faster. The assumption this architecture removes is that computation requires moving data to a separate processing unit at all. Photonic in-memory tensor cores and thermodynamic computing chips each independently demonstrate, in working silicon, that at least part of an AI workload's arithmetic can happen where the data already is, using physics rather than a fetch-execute-writeback cycle. Neither has been proven at model scale, neither has a software stack, and combining them is a research problem with no existing precedent. That is what makes it a genuine paradigm bet rather than a repackaged roadmap: the honest answer to 'has anyone shown this works' is no on the combination, yes on each half independently — and the roadmap in Section 5 is the specific, falsifiable sequence of experiments that would tell you, within a few years, whether the bet is paying off.

This document is an independent, speculative architectural synthesis prepared for research-planning purposes. It combines publicly reported early-stage research disclosures and does not represent the proprietary roadmap of any single vendor, nor a claim that the described system has been built, combined, or validated at scale.